

A Corrected Regular-Case Note on Generic Stochastic Subgradient Convergence Beyond Weak Convexity

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Abstract

Bianchi, Hachem, and Schechtman [2] proved that stochastic subgradient descent escapes active strict saddles and, generically in the definable weakly convex class, bounded stochastic subgradient trajectories converge to local minimizers. They also identified a beyond-weak-convexity obstruction: for nonregular locally Lipschitz definable functions, generic Clarke critical points may include sharply repulsive points, not only local minimizers and active strict saddles.

This note gives the corrected regular-case consequence. If the objective is locally Lipschitz, definable in an o-minimal structure, and Clarke/subdifferentially regular, then weak convexity is not needed: after a full-measure linear tilt, every bounded admissible stochastic subgradient trajectory converges almost surely to a local minimizer. The proof is not a new solution of the fully nonregular problem. It is a specialization of the regular-definable active-manifold theory of Davis, Drusvyatskiy, and Jiang [4], together with the tame stochastic approximation theorem of Davis, Drusvyatskiy, Kakade, and Lee [3]. Consequently the status of the result is *partial progress only*: it settles the regular definable subcase and leaves the sharply repulsive nonregular case untouched.

1 Status and scope

The theorem below is deliberately narrower than the full beyond-weak-convexity question. It proves

locally Lipschitz, definable, and subdifferentially regular

$\implies$

generic bounded stochastic subgradient convergence to local minima.

Thus weak convexity is removed, but regularity is added. This is a substantial restriction. It rules out the downward-cusp behavior responsible for sharply repulsive Clarke critical points. Accordingly, the theorem is best viewed as a verified regular-case corollary of known results, not as an independent solution of the Bianchi–Hachem–Schechtman problem in full generality.

The proof also repairs two common unsafe shortcuts. First, boundedness of a noisy recursion and the fact that $\alpha_k \rightarrow 0$ do not by themselves justify all asymptotic-pseudo-trajectory conclusions one might want. We instead use the tame stochastic subgradient theorem of Davis–Drusvyatskiy–Kakade–Lee [3]. Second, the reduction of a nonsmooth recursion near an active manifold to a noisy Riemannian gradient recursion is a theorem, not a formal projection argument. We import that theorem from Davis–Drusvyatskiy–Jiang [4].

2 Notation and stochastic model

Throughout, “definable” means definable in a fixed α -minimal expansion of the real field. Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be locally Lipschitz. We write $\partial^\circ f(x)$ for the Clarke subdifferential and

$$\text{crit } f := \{x \in \mathbb{R}^d : 0 \in \partial^\circ f(x)\}$$

for the Clarke critical set. A locally Lipschitz function is called regular if it is Clarke regular, equivalently subdifferentially regular in the standard locally Lipschitz variational-analytic sense. Linear tilts preserve local Lipschitzness, definability, and regularity. For $u \in \mathbb{R}^d$, set

$$f_u(x) := f(x) - \langle u, x \rangle.$$

We use the following stochastic subgradient model. It is intentionally matched to the uniform-noise strict-saddle avoidance theorem used below.

Assumption 2.1 (Admissible stochastic subgradient recursion). *Let $g: \mathbb{R}^d \rightarrow \mathbb{R}$ be locally Lipschitz. A process $(x_k)_{k \geq 0}$ is an admissible stochastic subgradient recursion for g if, on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_k), \mathbb{P})$, there is a fixed measurable selection $s: \mathbb{R}^d \rightarrow \mathbb{R}^d$ with $s(x) \in \partial^\circ g(x)$ for all x , and*

$$x_{k+1} = x_k - \alpha_k(s(x_k) + \xi_{k+1}), \quad (1)$$

where x_k is $\mathcal{F}_k$ -measurable, the steps satisfy

$$c_1(k+1)^{-\gamma} \leq \alpha_k \leq c_2(k+1)^{-\gamma} \quad \text{for some } c_1, c_2 > 0, \quad \gamma \in (1/2, 1), \quad (2)$$

and the noise variables ξ_{k+1} are independent of $\mathcal{F}_k$, centered, bounded, and nondegenerate. For concreteness, one may take the ξ_k independent and uniformly distributed on a fixed Euclidean ball centered at the origin. This is the stochastic subgradient method with fixed measurable selection considered in Davis–Drusvyatskiy–Jiang [4, Section 4.1 and Corollary 6.5].

The exact distributional assumption can be replaced by the equivalent nondegeneracy hypothesis in the strict-saddle avoidance theorem being invoked [4, Theorem 6.1]. What matters is bounded noise with a nontrivial component in every tangent direction of every active manifold. If one wants to remain completely literal to a particular published theorem, “admissible” should be read as “satisfying that theorem’s stated measurability, selection, step-size, and noise assumptions.”

3 Imported facts

We use two external theorems in specialized form. The first gives criticality of cluster points for bounded tame stochastic subgradient trajectories [3]. The second supplies the generic regular active-manifold structure and the stochastic avoidance of active strict saddles [4].

Theorem 3.1 (Tame stochastic criticality; Davis–Drusvyatskiy–Kakade–Lee). *Let $g: \mathbb{R}^d \rightarrow \mathbb{R}$ be locally Lipschitz and have Whitney-stratifiable graph; in particular, this holds for locally Lipschitz definable g . For any admissible stochastic subgradient recursion (1), on the event that (x_k) is bounded, every cluster point of (x_k) is Clarke critical for g .*

Theorem 3.2 (Generic regular active-manifold structure and avoidance; Davis–Drusvyatskiy–Jiang). *Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be locally Lipschitz, definable, and subdifferentially regular. Then there is a full-Lebesgue-measure set $U \subset \mathbb{R}^d$ such that, for every $u \in U$ and $g = f_u$, the following hold.*

- (i) The set $\text{crit } g$ is finite on bounded subsets of $\mathbb{R}^d$.
- (ii) Every critical point of g is either a local minimizer or an active strict saddle satisfying the active-manifold regularity, aiming, and tangential nondegeneracy hypotheses required by the stochastic avoidance theorem.
- (iii) For every active strict saddle $\bar{x}$ of g and every admissible stochastic subgradient recursion for g ,

$$\mathbb{P}(x_k \rightarrow \bar{x}) = 0.$$

Remark 3.3. Theorem 3.2 is the point at which regularity is used. In the regular definable class, generic critical points have the active-manifold structure needed for a Morse-type dichotomy: local minimum or active strict saddle. The fully nonregular class does not enjoy this dichotomy.

4 Corrected theorem

Theorem 4.1 (Regular definable generic convergence). *Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be locally Lipschitz, definable, and Clarke/subdifferentially regular. Then there exists a full-Lebesgue-measure set $U \subset \mathbb{R}^d$ such that for every $u \in U$, writing $g = f_u$, every admissible stochastic subgradient recursion (1) for g satisfies*

$$\mathbb{P}(\{x_k\} \text{ is bounded and } x_k \not\rightarrow \arg\min_{\text{loc}} g) = 0.$$

Equivalently, conditional on boundedness of the iterates, the sequence converges almost surely to a local minimizer of f_u .

Proof. Choose u in the full-measure set supplied by Theorem 3.2, and set $g = f_u$. Fix an admissible stochastic subgradient recursion for g .

For $R \in \mathbb{N}$, let

$$E_R := \left\{ \sup_{k \geq 0} \|x_k\| \leq R \right\}.$$

On E_R , local Lipschitzness of g implies that the selected subgradients are bounded, because the iterates remain in a compact set and Clarke subgradients are locally bounded for locally Lipschitz functions. The noise is bounded and $\alpha_k \rightarrow 0$, hence

$$\|x_{k+1} - x_k\| \rightarrow 0 \quad \text{on } E_R. \tag{3}$$

By Theorem 3.1, every cluster point of (x_k) on E_R lies in $\text{crit } g$. By Theorem 3.2(i), the set

$$C_R := \text{crit } g \cap \overline{B(0, R)}$$

is finite.

We use the following elementary fact: if a bounded sequence has vanishing increments, then its cluster set is connected. Indeed, if the cluster set were split into two nonempty closed pieces separated by a positive distance, then infinitely many transitions between small neighborhoods of the two pieces would force infinitely many iterates to lie outside both neighborhoods; compactness would then produce an additional cluster point outside the alleged split. Combining connectedness with the inclusion of the cluster set in the finite set C_R , we conclude that on E_R the sequence (x_k) converges to a single critical point of g .

By Theorem 3.2(ii), each point of C_R is either a local minimizer or an active strict saddle. For every active strict saddle $\bar{x} \in C_R$, Theorem 3.2(iii) gives

$$\mathbb{P}(x_k \rightarrow \bar{x}) = 0.$$

Since C_R is finite, the union over all nonminimizing critical points in C_R still has probability zero. Therefore

$$\mathbb{P}(E_R \cap \{x_k \text{ converges to a nonminimizing critical point}\}) = 0.$$

Finally,

$$\{x_k \text{ is bounded}\} = \bigcup_{R=1}^{\infty} E_R.$$

Taking the countable union over R proves the theorem. $\square$

5 Why this is not a full solution

The regularity hypothesis in Theorem 4.1 excludes the basic pathology that makes the fully nonregular problem difficult.

Example 5.1 (A nonregular sharply repulsive model). *Define*

$$q(y, z) = -|y| + |z| \quad ((y, z) \in \mathbb{R}^2).$$

At the origin,

$$\partial^\circ q(0, 0) = [-1, 1] \times [-1, 1],$$

so $(0, 0)$ is Clarke critical. It is not a local minimizer, since $q(t, 0) = -|t| < 0$ for $t \neq 0$. It is also not an active strict saddle of the regular active-manifold type used above. The ordinary directional derivative at the origin is

$$q'(0, 0; a, b) = -|a| + |b|,$$

whereas the Clarke directional derivative is

$$q^\circ(0, 0; a, b) = |a| + |b|.$$

These differ whenever $a \neq 0$. Thus q is not Clarke regular at the origin.

Example 5.1 is the same elementary obstruction emphasized by Bianchi–Hachem–Schechtman [2, Example 2 and Section 5.2], and shows why the corrected theorem should not be advertised as a resolution of their problem in full generality. The proof of Theorem 4.1 works because, after a generic tilt, regular definable critical points are exhausted by local minimizers and active strict saddles. In the nonregular definable class there may be sharply repulsive critical points, and avoiding them requires an additional argument not supplied here.

6 Verification checklist

The repaired argument differs from the initial manuscript in the following important ways.

- (1) **No unsupported asymptotic-pseudo-trajectory step.** Criticality of bounded stochastic subgradient trajectories is imported from the tame stochastic approximation theorem [3], rather than inferred only from boundedness and $\alpha_k \rightarrow 0$.

- (2) **No handwaved shadow recursion.** Avoidance of active strict saddles is imported from the active-manifold stochastic avoidance theorem [4, Theorem 6.2], where the projection to the active manifold, summable errors, aiming estimates, and nondegenerate tangential noise are proved.
- (3) **No random finite-union gap.** The proof works on deterministic bounded events E_R and then takes a countable union over R .
- (4) **Correct status.** The theorem is a regular-definable partial progress statement, essentially a corollary of existing results. It does not solve the fully nonregular sharply-repulsive case.

7 Conclusion

The repaired result is mathematically valid under the stated regularity and admissibility hypotheses: weak convexity can be removed for locally Lipschitz definable functions that are Clarke/subdifferentially regular. This is partial progress only. It is not an independent solution of the broader Bianchi–Hachem–Schechtman beyond-weak-convexity problem, because that problem requires dealing with nonregular sharply repulsive critical points.

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